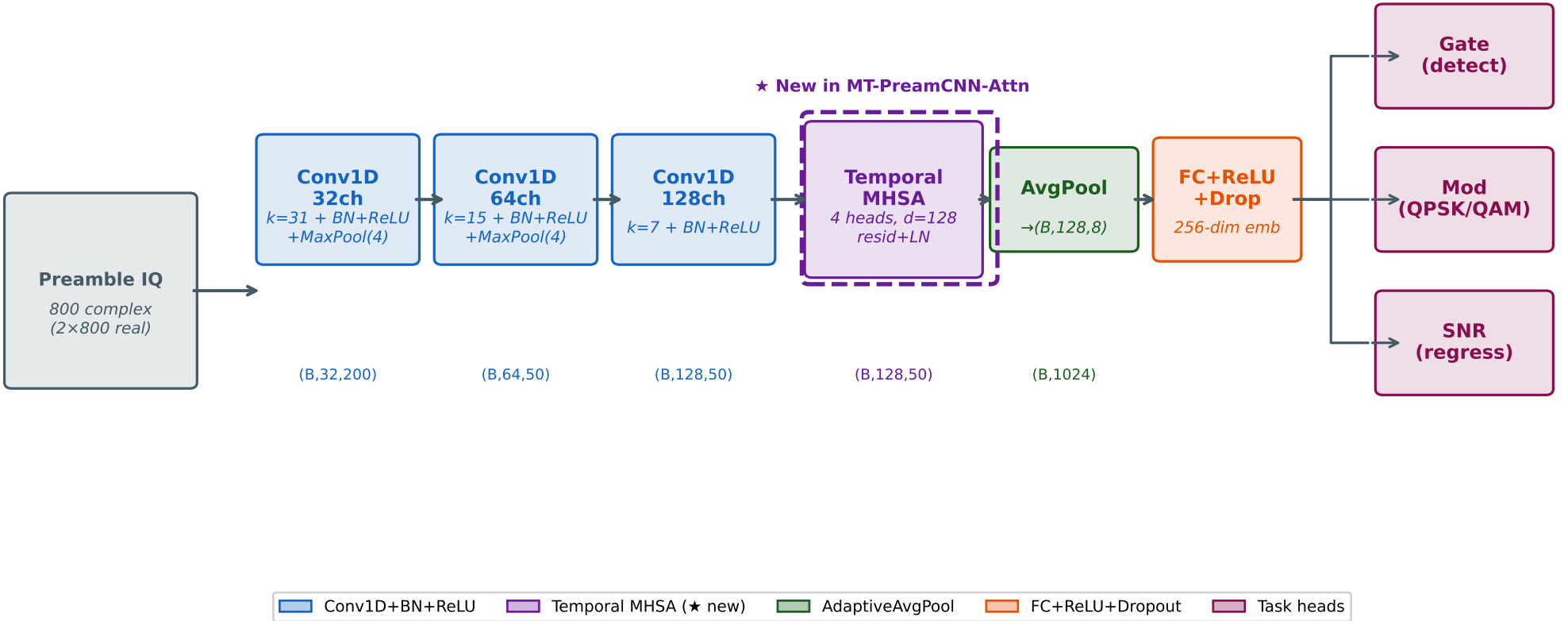
MT-PreamCNN-Attn Architecture
Shared CNN Encoder + Temporal Self-Attention → 3 Task Heads
★ New in MT-PreamCNN-Attn
Preamble IQ
800 complex
(2×800 real)
Conv1D
32ch
k=31 + BN+ReLU
+MaxPool(4)
Conv1D
64ch
k=15 + BN+ReLU
+MaxPool(4)
Conv1D
128ch
k=7 + BN+ReLU
Temporal
MHSA
4 heads, d=128
resid+LN
AvgPool
→(B,128,8)
FC+ReLU
+Drop
256-dim emb
Gate
(detect)
Mod
(QPSK/QAM)
SNR
(regress)
(B,32,200)
(B,64,50)
(B,128,50)
(B,128,50)
(B,1024)
Conv1D+BN+ReLU
Temporal MHSA (★ new)
AdaptiveAvgPool
FC+ReLU+Dropout
Task heads